A 7-vertex tournament whose Sperner capacity exceeds the order of its largest transitive subtournament

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Abstract

For a digraph R, let $w(R^n)$ be the largest set of words of length n in which every ordered pair of distinct words has a coordinate carrying an arc from the first to the second, and let $C(R) = \lim_n w(R^n)^{1/n}$ be the capacity. For a tournament T, $t(T) \le C(T)$, where $t(T)$ is the order of its largest transitive subtournament. The equality $C(T) = t(T)$, stated as Conjecture 1.1 in Alon's paper *On the capacity of digraphs* (1998) and attributed there to Körner and Simonyi, was disproved in the same paper; the smallest counterexample given there has 67 vertices, and Alon asked for the smallest size. We show that the answer is seven. The subtournament T_7 of the Paley tournament on 23 vertices induced on $\{0, 1, 2, 3, 9, 14, 18\}$ has $t(T_7) = 4$ and an 18-point transitive clique of its second Sperner power, so $\sqrt{18} \le C(T_7)$; a rank-5 matrix gives $C(T_7) \le 5$. Every tournament on at most six vertices has $C = t$, and on seven vertices T_7 is the only exception up to isomorphism. So T_7 is the unique smallest counterexample, and every proper subtournament of it has $C = t$. The statements about T_7 and the Paley tournament on 23 vertices, the general tools, and the classification of tournaments on at most six vertices are proved in Lean 4; the uniqueness on seven vertices and the exact value of the square are on paper, with exact certificates.

1 Definitions and the question

We follow Alon [1]. Let R be a digraph on a finite set V, written $u \to v$ for an arc. A set $S \subseteq V^n$ is a *clique of the n-th Sperner power* of R if for every ordered pair $u \ne v$ of members of S there is a coordinate i with $u_i \to v_i$. The pair is ordered: the coordinate for (u, v) and the coordinate for (v, u)

are in general different. Let $w(R^n)$ be the largest size of such a set. The function $n \mapsto w(R^n)$ is supermultiplicative, so the limit

$$C(R) = \lim_{n\to\infty} w(R^n)^{1/n} = \sup_{n\geq 1} w(R^n)^{1/n}$$

exists and equals the supremum. This is the *capacity* of R, also called its Sperner capacity [6]; some authors use the name for $\log C(R)$ [5, 4]. In Lean the capacity is defined as the supremum, so nothing about limits is assumed.

We also use a one-sided notion. A sequence $u^1, \ldots, u^s$ of distinct members of V^n is a *transitive clique* of the n-th Sperner power if for every $i < j$ some coordinate carries an arc from u^i to u^j. Let $\mathrm{tr}(R^n)$ be its largest length. Clearly $w(R^n) \leq \mathrm{tr}(R^n)$. For a tournament T and $u \neq v$, no coordinate carries an arc from u to v exactly when, in every coordinate, $v_i = u_i$ or $v_i \to u_i$, that is, when $v \to u$ in the directed strong power. So a transitive clique of the n-th Sperner power of a tournament is the same thing as an acyclic set of its n-th strong power, listed in a topological order.

A tournament T has exactly one arc between any two distinct vertices; it is transitive if its arcs follow a linear order. Let $t(T)$ be the order of the largest transitive subtournament of T. Then $t(T) \leq C(T)$ (Corollary 3 below).

Conjecture 1 (Conjecture 1.1 of [1])**.** *For every tournament T, $C(T) = t(T)$.*

Alon attributes the conjecture to Körner and Simonyi, citing [3]. The same paper disproves it: Theorem 1.3 there gives, for every n, a tournament on $2n$ vertices with $C \geq \sqrt{n}$ and $t \leq 4\log_2 n + 2$, and Section 4 there gives an explicit counterexample, the Paley tournament Q_{67}, which has no transitive subtournament on 9 vertices (a computer check reported there) while $C(Q_{67}) \geq \sqrt{67} > 8$. Section 4 of [1] adds that it may be interesting to determine the smallest size of a tournament for which the conjecture fails. On the small cases, [1] argues the conjecture in print for every tournament on at most 4 vertices, from its Theorem 1.2, and reports a verification, with Szabó and Tardos, for at most 5 vertices, with no proof printed. Kiviluoto, Östergård and Vaskelainen [6] determine the Sperner capacity of every digraph on at most 5 vertices except eight, none of which is a tournament. So before this note the least order of a counterexample was known to be at least 6.

Throughout, Q_p is the Paley tournament on $\mathbb{Z}/p$ for a prime $p \equiv 3 \pmod 4$: $x \to y$ if and only if $y - x$ is a nonzero square. In Lean the relation on $\mathbb{Z}/23$ is `paley`; the Lean name of each statement is given in brackets.

2 Two tools

Lemma 2 (the lift; `lift_lemma`)**.** *Let $p_0, \dots, p_{N-1} \in V^k$ be distinct, with an arc from p_i to p_j in some coordinate whenever $i < j$. Then for every $m \geq 1$ there is a clique S of the km-th Sperner power of R with $N^m \leq ((N-1)m+1)\,|S|$.*

Proof. For a word $a = (a_1, \dots, a_m) \in \{0, \dots, N-1\}^m$ let $L(a) \in V^{km}$ be the concatenation $p_{a_1} p_{a_2} \cdots p_{a_m}$. Let $a \neq b$ be two words with $\sum_q a_q = \sum_q b_q$. If $a_q \geq b_q$ for every q, the equal sums force $a = b$; so some q has $a_q < b_q$, and symmetrically some q' has $a_{q'} > b_{q'}$. In block q the point p_{a_q} sends an arc to p_{b_q} in some coordinate, so $L(a)$ sends an arc to $L(b)$ in the corresponding coordinate of V^{km}; block q' gives an arc from $L(b)$ to $L(a)$. Since the p_i are distinct, L is injective. So every class of words with a fixed index sum is mapped by L to a clique of the km-th power. The index sum takes the $(N-1)m+1$ values $0, \dots, (N-1)m$, so some class has at least $N^m/((N-1)m+1)$ words. □

The proof uses only the forward property for $i < j$. It never composes arcs, and it does not care whether some pairs also carry arcs backward. The lemma is standard in substance: the definition of the capacity through transitive cliques of the powers goes back to Gargano, Körner and Vaccaro [5], and Sali and Simonyi [11] prove the equivalence by a lift with the bound $\mathrm{tr}(R^k)^m/(m+1)^{\mathrm{tr}(R^k)}$ (see also [14, Section 2.2.1]); the equal-index-sum class gives the sharper elementary bound used here, which is the form proved in Lean.

Corollary 3 (`rpow_le_capacity`, `transitiveNumber_le_capacity`)**.** *With $p_0, \dots, p_{N-1}$ as in Lemma 2, $N^{1/k} \leq C(R)$. Hence $\mathrm{tr}(R^k)^{1/k} \leq C(R)$ for every k, and $t(R) \leq C(R)$.*

Proof. By the lemma, $C(R) \geq w(R^{km})^{1/(km)} \geq N^{1/k}\,((N-1)m+1)^{-1/(km)}$ for every m, and the last factor tends to 1. The second statement is the first for a longest transitive clique. The third is the case $k = 1$, with the vertices of a largest transitive subtournament in their order. □

Since $w \leq \mathrm{tr}$, the corollary shows $C(R) = \sup_k \mathrm{tr}(R^k)^{1/k}$ as well: defining the capacity through transitive cliques, as [6] do, gives the same number. A transitive clique of the k-th power with more than t^k points therefore proves $C > t$.

Lemma 4 (the rank bound; `capacity_le_of_factorization`)**.** *Let $\mathbb{F}$ be a field and $M \in \mathbb{F}^{V\times V}$ a matrix with $M_{vv} \neq 0$ for every v and $M_{uv} = 0$ whenever $u \to v$. If $r = \operatorname{rank} M$, then $\operatorname{tr}(R^n) \le r^n$ for every n; in particular $w(R^n) \le r^n$ and $C(R) \le r$.*

Proof. Let $K = M^{\otimes n}$, indexed by V^n: $K_{uv} = \prod_i M_{u_i v_i}$. Then $\operatorname{rank} K = r^n$; equivalently, if $M = AB$ with $A \in \mathbb{F}^{V\times r}$ and $B \in \mathbb{F}^{r\times V}$, then $K = A^{\otimes n}B^{\otimes n}$ factors through $\mathbb{F}^{r^n}$. Let $u^1, \dots, u^s$ be a transitive clique of the n-th power. For $i < j$ some coordinate c has $u^i_c \to u^j_c$, so $M_{u^i_c u^j_c} = 0$ and $K_{u^i u^j} = 0$; and $K_{u^i u^i} = \prod_c M_{u^i_c u^i_c} \neq 0$. So the $s \times s$ submatrix of K on these points is triangular with a nonzero diagonal, and $s \le \operatorname{rank} K = r^n$. For a clique the submatrix is diagonal. □

In Lean the hypothesis is stated as a factorization $M = AB$ through $\mathbb{F}^r$, and the conclusion is $C(R) \le r$. Alon's Theorem 1.2 [1], $C(D) \le d^- + 1$ for a digraph of maximum in-degree d^-, is the case $M_{uv} = \prod_{j\in N^-(v)}(u - j)$ over $\mathbb{Q}$ with $V = \{1, \dots, k\}$: as functions of u, the columns are polynomials of degree at most d^-, so $\operatorname{rank} M \le d^- + 1$. Reversing all arcs gives $C(D) \le d^+ + 1$. The lemma is the rank method of Haemers [7] as adapted to digraphs by Calderbank, Frankl, Graham, Li and Shepp [8] and by Blokhuis [9] (Alon [1] describes their method as a modification of Haemers'); for the Sperner capacity of digraphs it is stated, over the reals, as Theorem 4 of [6], who credit it to [8], and Alon's survey [2] gives an equivalent vector-representation bound over any field, credited to Alon and Tardos. We call the least rank of such a matrix over $\mathbb{F}$ the $\mathbb{F}$-*minrank* of R. For a tournament, $\operatorname{tr}(T^1) = t(T)$, so the lemma at $n = 1$ shows that the minrank is never below t; a matrix of rank t proves $C = t$.

Remark 5 (deletion). For any vertex v, $C(T) \le C(T - v) + 1$. Group a clique of T^n by the set A of coordinates equal to v; two words of one group agree on A, so their arcs lie outside A, and deleting A leaves a clique of $(T - v)^{n-|A|}$. Hence $w(T^n) \le \sum_k \binom{n}{k} w((T - v)^{n-k}) \le (C(T - v) + 1)^n$. We use this only to compare bounds.

3 The tournament T_7

Let $T_7 = Q_{23}[W]$ with $W = \{0, 1, 2, 3, 9, 14, 18\}$ [`T7`]. The nonzero squares modulo 23 are $1, 2, 3, 4, 6, 8, 9, 12, 13, 16, 18$, and -1 is not a square, so T_7 is a tournament [`T7_isTournament`]. Its 21 arcs are

$$0\to1,\ 0\to2,\ 0\to3,\ 0\to9,\ 0\to18,\ 1\to2,\ 1\to3,\ 1\to9,\ 1\to14,\ 2\to3,\ 2\to14,$$
$$2\to18,\ 3\to9,\ 9\to2,\ 9\to18,\ 14\to0,\ 14\to3,\ 14\to9,\ 14\to18,\ 18\to1,\ 18\to3.$$

The out-degrees of $0, 1, 2, 3, 9, 14, 18$ are $5, 4, 3, 1, 2, 4, 2$, so the score sequence is $(1, 2, 2, 3, 4, 4, 5)$, and the maximum out- and in-degrees are both 5.

Proposition 6 (`transitiveNumber_T7`)**.** $t(T_7) = 4$.

Proof. $0, 1, 2, 3$ is transitive in this order. No five vertices are transitive: a transitive subtournament grows only through common out-neighbours of the vertices chosen so far, and the search over such chains finds none of length 5. The kernel runs the same pruned search. □

Theorem 7 (`T7_square_witness`, `sqrt18_le_capacity_T7`)**.** *The eighteen points*

$$(0,0),\ (1,0),\ (14,1),\ (0,1),\ (0,9),\ (1,9),\ (14,2),\ (0,2),\ (9,14),$$
$$(2,14),\ (3,0),\ (9,0),\ (9,3),\ (2,3),\ (3,9),\ (9,9),\ (18,2),\ (18,18)$$

of $W \times W$, in this order, form a transitive clique of the second Sperner power of T_7. Hence $\sqrt{18} \le C(T_7)$, and in particular $t(T_7) = 4 < C(T_7)$ [`four_lt_capacity_T7`, `transitiveNumber_lt_capacity_T7`].

Proof. The forward property is a check of the 153 pairs (by `decide` in the kernel). Then Corollary 3 with $k = 2$ and $N = 18$ gives $\sqrt{18} \approx 4.243 \le C(T_7)$. □

The lift makes the counterexample concrete. With $N = 18$, $k = 2$ and $t^2 = 16$: at $m = 59$ one has $16^{59}\,(17 \cdot 59 + 1) < 18^{59}$, so the largest index-sum class gives $w(T_7^{118}) > 4^{118}$ [`T7_exists_clique_gt_four_pow`]; and the largest coefficient of $(1 + x + \cdots + x^{17})^{38}$ already exceeds 16^{38}, so $w(T_7^{76}) > 4^{76}$. Both are exact integer comparisons (certificate 0001).

Proposition 8. $\mathrm{tr}(T_7^2) = 18$. *There are exactly* 84 *transitive cliques of size* 18 *in the second Sperner power of T_7.*

The lower bound is Theorem 7. The upper bound is proved by the exhaustive search of certificate 0003, a standard-library checker that runs in about ten minutes; it was also obtained by three engines that share no code (a CP-SAT model with positions, a branch and bound in pure Python, and a SAT model with lazy cycle cuts solved by CaDiCaL). The count of 84 is a computation: two of the engines return identical lists. The 17-point set found first in this work is inclusion-maximal but not maximum: no point extends it, and it lies in none of the 84 optimal sets.

Theorem 9 (`capacity_T7_le_five`)**.** $C(T_7) \le 5$.

Proof. Index rows and columns by $0, 1, 2, 3, 9, 14, 18$ and let

$$M = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & -1 & 0 & 1 \end{pmatrix},$$

so that the rows are e_0, $e_1 + e_{18}$, $e_1 + e_2 + e_9$, $-e_2 + e_3 + e_{18}$, $e_3 + e_9$, e_{14} and $-e_2 - e_9 + e_{18}$. The diagonal is 1 and every arc position is 0 (compare with the list of arcs). The rows satisfy $v_{18} = v_1 - v_2 = v_3 - v_9$, and the rows at $0, 1, 2, 3, 14$ are independent, so $\operatorname{rank}_{\mathbb{Q}} M = 5$; the two relations come from the cyclic triangles $1 \to 2 \to 18 \to 1$ and $3 \to 9 \to 18 \to 3$. Over $\mathbb{F}_2$ the same matrix factors as $M = AB$ with

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}:$$

the rows of B are the rows of M at $0, 1, 2, 3, 14$, and the rows at 9 and 18 are $v_1 + v_2 + v_3$ and $v_1 + v_2$. Lemma 4 with $r = 5$ gives the claim. □

Alon's degree bound gives only $C(T_7) \leq 6$. Rank 4 is impossible over every field, since Lemma 4 would give $\operatorname{tr}(T_7^2) \leq 16 < 18$; exhaustive searches over $\mathbb{F}_2$ and $\mathbb{F}_3$ agree. So

$$4.2426\ldots = \sqrt{18} \leq C(T_7) \leq 5.$$

Proposition 10. *$w(T_7^2) = 7$, and the only clique of size 7 in the second Sperner power is the graph of the anti-automorphism $\varphi = (0\ 3)(1\ 18)(9\ 14)$, which fixes 2.*

Proof. If two members (a, b), (a, b') of a clique share a coordinate, both arcs between them must come from the other coordinate, so $b \to b'$ and $b' \to b$, which is impossible. So a clique of the second power is the graph of a partial injection and has at most 7 points. A 7-point clique is the graph $\{(a, \pi(a))\}$

of a bijection π, and for $a \to a'$ the pair needs $\pi(a') \to \pi(a)$: π is an anti-automorphism. Conversely the graph of an anti-automorphism is a clique. A search over all 7! permutations finds $\mathrm{Aut}(T_7)$ trivial and exactly one anti-automorphism, φ; it is unique because the anti-automorphisms form a coset of $\mathrm{Aut}(T_7)$. □

So T_7 is self-converse with a trivial automorphism group, and the square gives a clique of only 7 points: the separation from t^n appears for transitive cliques at $n = 2$, and for cliques only in a high power. The following are computations, labelled as such.

- $75 \le \mathrm{tr}(T_7^3) \le 125$. The lower bound is a witness found by simulated annealing and checked exactly (certificate 0007); the upper bound is Lemma 4. Since $75^2 < 18^3$, the cube does not improve $\sqrt{18}$; a value of at least 77 would.
- $16 \le w(T_7^3) \le 19$. The lower bound is a checked witness; the upper bound is a SAT refutation of 20 by two solvers (CaDiCaL and Glucose, with different encodings).
- Fixing one coordinate shows $w(T^n) \le |V|\, w(T^{n-1})$, so $w(T_7^4) \le 133 < 4^4$ and $w(T_7^5) \le 931 < 4^5$. With $w(T_7) = 1$, $w(T_7^2) = 7$ and $w(T_7^3) \le 19 < 4^3$, no clique of T_7^n exceeds 4^n for $n \le 5$, and the least n with $w(T_7^n) > 4^n$ lies in $[6, 76]$.

4 Tournaments on at most six vertices

Theorem 11 (`capacity_eq_transitiveNumber_of_card_le_six`)**.** *Every tournament T on at most six vertices has $C(T) = t(T)$.*

Proof. There are 1, 1, 2, 4, 12 and 56 isomorphism classes of tournaments on one to six vertices. For each class we list a representative T and an $\mathbb{F}_2$-matrix with diagonal 1, zeros on the arcs and rank $t(T)$ (certificate 0004). Lemma 4 gives $C(T) \le t(T)$, and Corollary 3 gives $C(T) \ge t(T)$. The list of representatives is complete: the representatives are pairwise non-isomorphic, and $\sum n!/|\mathrm{Aut}(T)| = 2^{n(n-1)/2}$, the number of labelled tournaments on n vertices, for each n. The checker recomputes t, the zero pattern, the ranks over $\mathbb{F}_2$ and both completeness conditions.

The theorem is also proved in Lean, by a different completeness argument that the kernel checks: a tournament on n vertices is encoded by its $\binom{n}{2}$ orientation bits; for each $n \le 6$ the representatives, with a factorization

AB over $\mathbb{F}_2$ of rank t and a transitive t-chain each, are listed, and the kernel verifies the data and that the bitwise *or* of 2^{code} over every representative and every one of the $n!$ relabellings equals $2^{2^{\binom{n}{2}}} - 1$, so that every orientation is a relabelled representative; the factorization and the chain are then transported along the relabelling. For $n = 6$ this is $56 \cdot 720$ relabellings and about a minute of kernel time. □

For example, the 6-vertex class obtained from Q_7 by deleting a vertex has $t = 3$, degree bound 4, and an $\mathbb{F}_2$-matrix of rank 3. An exhaustive search over all admissible $\mathbb{F}_2$-matrices (the $\binom{n}{2}$ entries opposite the arcs are free) shows more: *the $\mathbb{F}_2$-minrank equals t for every tournament on at most six vertices.* This is a computation, not needed for the theorem.

Corollary 12 (`seven_le_card_of_transitiveNumber_lt_capacity`)**.** *The least order of a tournament T with $C(T) > t(T)$ is 7.*

Proof. Theorem 11 and Theorem 7. □

As a consistency check, two further engines computed $\mathrm{tr}(T^2)$ for every class on at most six vertices; it equals t^2 in every case, as Lemma 4 requires.

5 Seven vertices

Theorem 13. *Of the 456 isomorphism classes of tournaments on seven vertices, 455 have an $\mathbb{F}_2$-matrix with diagonal 1, zeros on the arcs and rank t, hence $C = t$. The remaining class is that of T_7.*

This is certificate 0005: the 456 representatives, complete by the same orbit count ($\sum 7!/|\operatorname{Aut}(T)| = 2^{21}$), 455 matrices, and an explicit isomorphism from the remaining representative to T_7. The values of t over the 456 classes are $3, 4, 5, 6, 7$ with multiplicities $1, 120, 287, 47, 1$.

Corollary 14. *Up to isomorphism, T_7 is the only tournament on seven vertices with $C > t$, and it is the unique smallest tournament with $C > t$. Since T_7 is self-converse, it is also unique up to reversing all arcs.*

Corollary 15. *Every proper subtournament T' of T_7 has $C(T') = t(T')$. The seven subtournaments on six vertices fall into five isomorphism classes, all with $t = 4$. So T_7 is vertex-minimal with $C > t$.*

Proof. Every proper subtournament has at most six vertices; apply Theorem 11. □

Körner asked ([1], Section 4) for a characterisation of the tournaments T in which every subtournament T' has $C(T') = t(T')$. Corollaries 14 and 15 answer this for tournaments on at most seven vertices: such a tournament has the property if and only if it is not isomorphic to T_7. Any tournament containing a copy of T_7 lacks it.

Proposition 16. *$C(Q_7) = 3$.*

Proof. Q_7 has arcs $x \to y$ for $y - x \in \{1, 2, 4\}$, and $0, 1, 2$ is transitive, so $t(Q_7) \geq 3$. Over $\mathbb{F}_2$ let $M_{uv} = 1$ if $u = v$ or $v \to u$, and 0 otherwise: the identity plus the converse adjacency matrix. It is the circulant of $f(x) = 1 + x^3 + x^5 + x^6$, and f is divisible by $(x+1)(x^3 + x + 1)$, a factor of $x^7 - 1$ of degree 4, so $\operatorname{rank}_{\mathbb{F}_2} M \leq 7 - 4 = 3$. Lemma 4 gives $C(Q_7) \leq 3$, and Corollary 3 gives $t(Q_7) = 3 = C(Q_7)$. □

Here the degree bound and the deletion bound both give only 4. Q_7 is the only tournament on seven vertices with $t = 3$. We have not searched the literature for this value beyond the sources of Section 8.

6 The Paley tournament on 23 vertices and its neighbours

T_7 was found inside Q_{23}, which is itself a counterexample with $t = 5$.

Theorem 17. *$t(Q_{23}) = 5$, and the 29 points*

$$(22,7),\ (8,2),\ (8,3),\ (8,11),\ (1,13),\ (14,13),\ (1,6),\ (5,6),$$
$$(17,22),\ (17,7),\ (3,22),\ (3,7),\ (12,19),\ (12,22),\ (1,15),\ (5,15),$$
$$(21,19),\ (14,2),\ (14,15),\ (9,0),\ (9,2),\ (17,0),\ (17,2),\ (7,4),$$
$$(3,8),\ (7,6),\ (9,8),\ (21,8),\ (11,8)$$

form, in this order, a transitive clique of the second Sperner power of Q_{23}. Hence $\sqrt{29} \leq C(Q_{23}) \leq 12$, and $5 < C(Q_{23})$.

Proof. $0, 1, 2, 3, 4$ is transitive. By translation, a transitive 6-set may be assumed to start at 0, and the pruned search over common out-neighbourhoods finds none (the value $t(Q_{23}) = 5$ is in print: [10], as reported in [13], and [12, Table 2]). The forward property is a check of 406 pairs, and Corollary 3 gives the lower bound. Q_{23} is regular of degree 11, so Alon's Theorem 1.2 gives the upper bound 12. □

The Lean names are `paley_isTournament`, `transitiveNumber_paley`, `five_lt_capacity_paley` and `sqrt29_le_capacity_paley`; the upper bound 12 is not formalized.

Alon's anti-diagonal $\{(x, -x)\}$ gives only $w(Q_{23}^2) \geq 23$, that is $C(Q_{23}) \geq \sqrt{23}$, which is below 5. The lift of the 29 points gives $w(Q_{23}^{98}) > 5^{98}$ by pigeonhole at $m = 49$ (certificate 0002).

Two examples smaller than Q_{67} follow from what is in print, although we did not find them stated. The Paley tournament ST_{27} on the field with 27 elements has $t = 5$ by Sanchez-Flores [10], and Alon's anti-diagonal $\{(a, -a)\}$, a clique of the second Sperner power because $x \mapsto -x$ reverses every arc, has 27 points; so $C(ST_{27}) \geq \sqrt{27} > 5$. Removing the point $a = 0$ gives a 26-vertex subtournament with $t \leq 5$ and a 26-point clique, so $C \geq \sqrt{26} > 5$. We re-verified $t = 5$ and both cliques (certificate 0008).

7 What is proved in Lean, and what is on paper

Nineteen theorems are stated in the file `Challenge.lean` of the repository `sperner-t7-lean` and proved in `Solution.lean`, checked by the Lean 4 kernel [15] against Mathlib [16] at tag v4.35.0-rc2, with no axioms beyond `propext`, `Classical.choice` and `Quot.sound`. The definitions are `paley`, `W`, `T7`, `IsTournament`, `IsTransitiveChain`, `transitiveNumber`, `IsSpernerClique`, `spernerCliqueNumber` and `capacity` (the supremum over $n \geq 1$ of $w(R^n)^{1/n}$). The statements are:

- about T_7: it is a tournament; $t(T_7) = 4$; the 18-point transitive clique of the second Sperner power; some power T_7^n has a clique larger than 4^n; $4 < C(T_7)$; $\sqrt{18} \leq C(T_7)$; $C(T_7) \leq 5$; $t(T_7) < C(T_7)$;
- general tools, for any digraph on a finite set: the lift (Lemma 2, in natural numbers, with no finiteness assumption); $N^{1/k} \leq C(R)$; $t(R) \leq C(R)$; the rank bound (Lemma 4, for a factorization $M = AB$ through $\mathbb{F}^r$ over any field); $C(R) \leq |V|$;
- about Q_{23}: it is a tournament; $t(Q_{23}) = 5$; $5 < C(Q_{23})$; $\sqrt{29} \leq C(Q_{23})$;
- about small tournaments: every tournament on at most six vertices has $C = t$ (Theorem 11), and a tournament with $t < C$ has at least seven vertices.

Theorem 13, Propositions 8, 10 and 16, the bounds on T_7^3, and the 26- and 27-vertex examples are on paper. Their finite data are checked by standalone Python programs in `note/certificates/` (standard library only, exact integers, no `assert` statements), each with forged inputs that must fail:

0001 the 18-point transitive clique of T_7^2, $t(T_7) = 4$, and the lift at $n = 118$ and $n = 76$
0002 the 29-point transitive clique of Q_{23}^2, $t(Q_{23}) = 5$, and the lift
0003 $\mathrm{tr}(T_7^2) = 18$ exactly, by an exhaustive upper-bound search
0004 $C = t$ on at most six vertices: the class lists and the rank-t matrices
0005 the 455 rank-t matrices on seven vertices and the isomorphism to T_7
0006 the rank-5 matrix for T_7 over $\mathbb{Q}$ and over $\mathbb{F}_2$
0007 the witnesses for $\mathrm{tr}(T_7^3) \geq 75$ and $w(T_7^3) \geq 16$
0008 $t = 5$ for the Paley tournament on 27 elements and the anti-diagonal cliques

The folder's README gives the re-run line of each certificate. The exact value 18, the count 84, the SAT bound $w(T_7^3) \leq 19$ and the exhaustive minrank searches are computations, and are labelled as such where they are stated.

8 Prior art

As of 2026-10-08 (UTC), we found no tournament on fewer than 67 vertices exhibited in print with capacity above its transitive number, and no statement that every tournament on six vertices has $C = t$. We searched [1] and Alon's survey [2]; the abstract of [3]; [6], whose conclusion leaves six or more vertices to further work, and Vaskelainen's thesis, whose 14-vertex tournament concerns Banks and Slater sets, not capacity; Simonyi's DSc dissertation [14]; the arXiv papers on Sperner capacity (arXiv:math/0312086, 0809.1522, 1309.4930, 1406.0767, 1905.10483, 2312.09224); the Palomar registry; Hexagon; the openai/math repository; and the formal-conjectures repository, zbMATH and Google Scholar. The nearest examples are the 26- and 27-vertex tournaments of Section 6, which follow in one line from [1] and [10]. The two tools of Section 2 are standard in substance ([11], [6]); their values on T_7 and the classification of the small tournaments are what is new.

9 Open questions

1. The exact value of $C(T_7)$, which lies in $[\sqrt{18}, 5]$. A transitive clique of the cube with 77 points would raise the lower bound.
2. The exact values of $\mathrm{tr}(T_7^3) \in [75, 125]$ and $w(T_7^3) \in [16, 19]$.
3. The least n with $w(T_7^n) > 4^n$, which lies in $[6, 76]$.
4. The least order of a counterexample with $t = 5$. By Section 5 it is at least 8, and Q_{23} shows it is at most 23. Computations made during this work, with a single implementation that we have not repeated, found induced subtournaments of Q_{23} on 8 to 11 vertices with a transitive clique of the second Sperner power exceeding t^2: 17 points for one 8-vertex example with $t = 4$, and 26 points for examples with $t = 5$ on 8, 9, 10 and 11 vertices. If confirmed, the answer is 8. We list these only as leads.
5. The capacity of Q_{23}, which lies in $[\sqrt{29}, 12]$, and of the Paley tournaments in general.

Provenance

Claude (Anthropic) was used throughout; the theorems, proofs, searches and checkers arose in that process, and the repository was written by Claude. The searches used OR-Tools CP-SAT and PySAT with the CaDiCaL and Glucose solvers; every claim that rests on a computation has a standard-library checker or is labelled as a computation. The Lean development was checked by the Lean 4 kernel [15] against Mathlib [16] at tag v4.35.0-rc2. The author directed the project and takes responsibility for the claims. This note and the repository are licensed MIT.

References

[1] N. Alon, On the capacity of digraphs, European J. Combin. 19 (1998) 1–5, doi:10.1006/eujc.1997.0148.

[2] N. Alon, Graph powers, in: Contemporary Combinatorics, Bolyai Soc. Math. Stud. 10, Springer (2002) 11–28.

[3] J. Körner, On clique growth in products of directed graphs, Graphs Combin. 14 (1998) 25–36, doi:10.1007/s003730050012.

[4] J. Körner, G. Simonyi, A Sperner-type theorem and qualitative independence, J. Combin. Theory Ser. A 59 (1992) 90–103, doi:10.1016/0097-3165(92)90100-9.

[5] L. Gargano, J. Körner, U. Vaccaro, Qualitative independence and Sperner problems for directed graphs, J. Combin. Theory Ser. A 61 (1992) 173–192, doi:10.1016/0097-3165(92)90016-N.

[6] J. Kiviluoto, P. R. J. Östergård, V. P. Vaskelainen, Sperner capacity of small digraphs, Adv. Math. Commun. 3 (2009), no. 2, 125–133, doi:10.3934/amc.2009.3.125.

[7] W. Haemers, On some problems of Lovász concerning the Shannon capacity of a graph, IEEE Trans. Inform. Theory 25 (1979) 231–232, doi:10.1109/TIT.1979.1056027.

[8] A. R. Calderbank, P. Frankl, R. L. Graham, W.-C. W. Li, L. A. Shepp, The Sperner capacity of linear and nonlinear codes for the cyclic triangle, J. Algebraic Combin. 2 (1993) 31–48, doi:10.1023/A:1022424630332.

[9] A. Blokhuis, On the Sperner capacity of the cyclic triangle, J. Algebraic Combin. 2 (1993) 123–124, doi:10.1023/A:1022455407000.

[10] A. Sanchez-Flores, On tournaments free of large transitive subtournaments, Graphs Combin. 14 (1998) 181–200, doi:10.1007/s003730050025.

[11] A. Sali, G. Simonyi, Orientations of self-complementary graphs and the relation of Sperner and Shannon capacities, European J. Combin. 20 (1999) 93–99, doi:10.1006/eujc.1998.0256.

[12] K. Momihara, S. Suda, Upper bounds on the size of transitive subtournaments in digraphs, Linear Algebra Appl. 530 (2017) 230–243, doi:10.1016/j.laa.2017.04.030.

[13] D. Neiman, J. Mackey, M. Heule, Tighter bounds on directed Ramsey number $R(7)$, Graphs Combin. 38 (2022), article 156, doi:10.1007/s00373-022-02560-5.

[14] G. Simonyi, Entropies, capacities and colorings of graphs, DSc dissertation, Hungarian Academy of Sciences (2006).

[15] L. de Moura, S. Ullrich, The Lean 4 theorem prover and programming language, CADE 28, Lecture Notes in Computer Science, Springer (2021) 625–635, doi:10.1007/978-3-030-79876-5_37.

[16] The mathlib Community, The Lean mathematical library, CPP 2020, 367–381, doi:10.1145/3372885.3373824.